JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY KAKINADA

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

ANALOG ELECTRONICS LABORATORY

★ HARDWARE EXPERIMENT LOGBOOK & MULTISIM GUIDELINES ★

Electronic Devices and Circuits (EDC) & Simulation Lab Manual

Course Code:	R23-ECE-LAB1	**Academic Scheme:**	I B.Tech I Semester (R23 Scheme)
Verification:	Verified Authentic Academic Copy	**System Ref:**	STUDY-BLUEPRINT-VERIFIED-2026

DETAILED CURRICULAR MODULES & TOPICS

Experiment 1: PN Junction Diode V-I Characteristics

Plotting forward and reverse characteristics, determining cut-in voltage and dynamic resistance.

Experiment 2: Zener Diode as a Voltage Regulator

Line regulation and load regulation characteristics, Breakdown voltage determination.

Experiment 3: Half Wave & Full Wave Rectifiers with Filters

Measuring ripple factor and DC output voltage with and without capacitor filters on DSO.

Experiment 4: Common Emitter (CE) Transistor Characteristics

Measuring input and output characteristics, Calculating h-parameters (hfe, hie, hoe, hre).

Experiment 5: JFET Drain & Transfer Characteristics

Determining transconductance gm, drain resistance rd, and amplification factor mu.

EVALUATION SCHEME & EXAMINATION BLUEPRINT

Lab Day-to-Day Grade	15 Marks
Practical Exam	50 Marks (Total: 75 Marks)

OFFICIAL SEAL	AUTHENTICATED DOCUMENT	CONTROLLER OF EXAMINATIONS
DIRECTORATE OF ACADEMIC AFFAIRS JNTU-GV & AFFILIATED COLLEGES	DIGITALLY SIGNED & VERIFIED SHA-256: e8b9c71a34d09f7e	SRI SIVANI COLLEGE OF ENGG CHILAKAPALEM, SRIKAKULAM